



A Tale of Two Plants: NUMMI Teamwork Versus GM Bureaucracy

One might not recognize Rick Madrid as the same man who carried, by his own admission, "a pen and a punch" as an inspector two years ago. Oh, the outward appearance hasn't changed much—tattoos and mirrored shades continue to be his trademarks. But the "Iron Maiden" T-shirt is now cleverly concealed beneath his sharply pressed shop coat provided by the company. He's a team leader now, qualified and sanctioned by his peers. He prefers to be called a team motivator, however. "I'm *part* of the team; I don't *have* a team. Let people maintain their own personality."

—GM Today publication

For those who believed that the Japanese industrial edge rested solely in technological prowess, the NUMMI experiment was a real revelation. The Toyota secret was, finally, no secret at all, and it was as old as history: Treat both white- and blue-collar workers with respect, encourage them to think independently, allow them to make decisions, and make them feel connected to an important effort. Combine that culture with a good car and quality parts, and the results are obvious. As a GMer who spent three years there said, "It was common sense." Going to work for NUMMI was a shock to the system of GM employees, who were accustomed to a stifling bureaucracy and an emphasis on high-tech solutions over worker initiative. At NUMMI, Toyota would demonstrate a business strategy based on trust, respect, and teamwork.

American companies tend, fundamentally, to mistrust workers, whether they are salaried employees or blue-collar

workers. There is a pervading attitude that "if you give them an inch, they'll take a mile," because they don't really want to work. The idea, for example, that a worker in the plant would have the power to stop the line in order to eliminate a problem was heresy. Wouldn't such permission lead to widespread line-stoppage for every whim? Not, according to Toyota, if you instituted a system of worker responsibility and accountability. American companies might think they had a tight rein on employees, but with little responsibility resting directly in the workers' laps, there was also little accountability.

And what were the workers being held responsible for? In the first instance, *quality*. In Japanese companies, quality was part of the process, not something added on in the inspection phase. American industry focused on "quality control." Generations of business-school graduates had learned that high quality equaled high cost. GM was not different from the rest of American industry in using inspectors to find defects *after* the fact. At General Motors, the way the company demonstrated to the world its concern for quality was to add more inspectors, to check for defects. The Japanese eliminated defects at all stages, from raw materials, to components, to assembly. The workers, not the inspectors, had the primary responsibility for the quality of their work, and they were given the tools to help them do a good job.

This is the tale of two plants, both in California. Fremont, the location of the NUMMI joint venture, would demonstrate that Japanese techniques can work in an unlikely environment. A few hundred miles away at the Van Nuys plant, GM was building cars the old-fashioned way, operating from very different assumptions about worker productivity, chain of command, and quality control.

TRADITION: THE GM PLANT IN VAN NUYS

Van Nuys was considered to be one of GM's more troubled plants, particularly because of its union problems. That's one of the reasons Brian Haun requested an assignment there as production supervisor. Haun was an engineer who had spent time in central staff offices, and he felt the Van Nuys experience would be valuable for his career.

Van Nuys built Pontiac Firebirds and Chevrolet Camaros. When Haun arrived, he was told that he would receive six months of training. He soon learned the nature of his training: *on the job*. And his trainers were the very people he was responsible for supervising.

On his first day, Haun was told to report to Section 25. There he introduced himself to the emergency relief operator, who was the head operator. "My name is Brian Haun and I'm your new supervisor," he told him. "I don't know anything about this." Fortunately the man taught Haun what he needed to know. "But I knew what he was thinking—what everyone was thinking. It was, 'Here comes a young college kid. Let's get him.'"

Haun was the butt of many jokes and the victim of what amounted to a plant-style fraternity hazing. He took it good-naturedly, waiting for an opportunity to show them he was on their side.

The opportunity came soon enough. Union workers were complaining about a problem with a defective welding robot. Whenever it went down, a repair operator had to pick up whatever welds the robot was missing. But the repair operator's job was limited contractually to doing repairs, not doing the welding that the robot was failing to do.

"But basically we were having the repair guy do the welding one hundred percent of the time," Haun said. "After about two weeks, he got fed up, and I thought he was right. So I went up to management and I tried to fight them on it, but they said, 'No. You make him do it, or else.'"

Haun refused, went back to his operators, and told them, "Stand your ground. Don't do it and we'll fight it out."

The union committeeman, the department superintendent, and a general supervisor who was Haun's boss argued heatedly. "They thought they could get the union to back down," Haun said. "But they couldn't, so finally, the hourly people won—and they should have. All of a sudden, I had the respect of everybody in the section. From then on, they broke their backs for me."

Despite learning his job in a confrontational environment, Haun gained a new perspective from his early days at the plant. He could see that the workers wanted to do their jobs well, wanted to be competitive, but all too often they were fighting against unbeatable odds to get their jobs done. Every problem

became a confrontation; since there was a basic mistrust between labor and management, it was hard to establish a cooperative environment where problems could be solved.

Haun also found that the absence of personal responsibility and accountability could lead to costly problems.

"We had a problem with a little bracket that gets welded on the back of the car," he said, citing an example. "It holds the sunshade in place. There was a man and a woman on each side of the car who put that bracket on, but they kept missing them. The car would reach the end of the production line and have to be repaired in the 'backyard' area where defects are repaired.

"The repair man would have to take all the carpet out and all the trim out and weld that bracket in—sometimes the carpet got burned. And then it all had to be put back together and, frankly, the car was never the same. So it's a bad deal."

Haun took the two workers off the production line and asked them to go with him to the backyard. Taking workers off the line was rarely done—it was too disruptive. But Haun decided that the disruption would be worth it if it resulted in more cars surviving the gauntlet of line workers without being damaged.

"I took them out and said, 'Look, this is what happens when you miss one of those,' said Haun. "The repair guy showed them how he had to rip out all the carpet, and they were shocked. And the woman said, 'You mean to tell me that bracket holds the sunshade?' She'd been doing this job for two years and nobody had ever told her what part she was welding."

This wasn't an isolated incident. Haun found that workers operated in a vacuum; they did the job they were told to do without relating it to the final outcome. "Basically, they didn't know who their immediate customers were. Their immediate customers were the guys who had to repair a defect they created, or the people down the line. And they didn't really know how much of an impact they personally had on the way the car was built. In the plant, you're treated like a no-mind idiot. You're supposed to come in and just put the parts on the car and shut up and do your job, and if you miss one, I'm going to yell at you, and eventually you get so used to the yelling that it doesn't do any good. If you're a supervisor, what can you do? You can't fire them—the union won't let you. I think you have to motivate them by positive means. Taking those two workers out into the

backyard, for example, really worked well. The woman is fantastic. She turned out to be one of the best operators I had."

Haun said that when he returned that day from showing the two operators the problems they had been causing, "I got in trouble with my general supervisor because they were short on manpower and they didn't want to pick people off the line. He said, 'We don't have enough people for you to do that. I needed those people over at Section Thirty.' But I knew I had to do it."

What Haun encountered was an environment where workers were not partners in the task of building cars. Often they didn't know how their jobs related to the total picture. Not knowing, there was no incentive to strive for quality—what did quality even mean as it related to a bracket whose function you didn't understand? Workers were held accountable through a system of intimidation: Do your job and your supervisor won't yell at you. That was a pretty thin incentive!

It is easy to understand why such an environment would produce workers who do little without being ordered to do it. Why should they? There is no reward for putting in the extra effort. And Van Nuys wasn't unique. It was typical.

In the past, Haun had observed workers on the third (night) shift of stamping plants spending most of the night just moving around aimlessly. "Six o'clock in the morning would come around and they wouldn't have the line ready, so they would have to rush at the last minute," he said.

It was part of a vicious circle that ultimately hurt GM where it could least afford to be hurt—in the area of quality. Because of the changeover delay caused by unmotivated workers, the press time was cut. "Then," explained Haun, "we didn't have time to have the dies repaired the way they should have been. Because they weren't getting repaired in the proper way, they'd just have quick fixes on them and the quality would suffer. You'd get what they call a 'ding.' It's a small deformity in part of the die, or something gets stuck in part of the die and you get a tiny dent that you can hardly see until the car is painted. Or, worse, you lose your dimensional control. For example, when you lose dimensional control on the door, it gets twisted and then it's really hard to fit the parts together in the assembly plant."

These are the kinds of small—often minuscule—defects that turn into major problems on the showroom floor. The customer wants a perfect car. He isn't going to say, "No prob-

lem. That little dent isn't important." What Haun observed was that the traditional operating attitudes about how workers should be treated were directly responsible for the quality problems that lessened GM's competitive advantage.

The prevailing attitude encouraged an adversarial relationship between workers and management. John McNulty, GM's vice president of public relations, was concise and brutally honest in describing the us-versus-them mentality.

"We are not yet a classless society," McNulty said. "We're not in the same adversary position we used to be in with the union, but fundamentally the mission of their elected representatives is to get the most compensation for the least amount of labor. Our responsibility to our stockholders is to get the most production for the least amount of compensation. And we do it in ways that are not just taking money away from the employees—with better technology, for example."

McNulty's remark reveals the philosophy that, more than anything, has contributed to GM's loss of the competitive edge. There is no trust. No respect. Instead, it's a fight: Control the hourly workers before they can control the company, and replace them with machines whenever possible.

REVELATION: NUMMI, FREMONT, CALIFORNIA

The Fremont plant, which would be the site of the joint venture, had been closed by General Motors in 1982, and five thousand employees were laid off. When GM and Toyota first discussed Fremont, Toyota was resistant to the idea of hiring from the old work force. It was notorious as being one of GM's most unmanageable. Before the plant was closed, daily absenteeism was regularly over 20 percent; beer bottles littered the parking lot; and even the slightest dispute had to go to the bargaining table. But Toyota and GM worked together during the selection process and managed to hire workers from the original worker pool—avoiding those who were known to be troublemakers.

"I was determined that it would be the Fremont workers, and that took some hard bargaining because Toyota didn't want the Fremont workers," said Al Warren, vice president of labor relations for GM. "I'm glad they did it, because it demonstrated that a work force we thought was unmanageable could be man-

aged. That was a tremendous process. They took almost everybody back—I don't think there was more than a handful that didn't get back eventually, but they did a masterful job of that."

In negotiating with the UAW, Toyota agreed to give the union a strong job security clause and a greater say in plant operations. In return, the union agreed to work within Toyota's system, eliminating unproductive job classifications, and setting up a team assembly system.

The secret of NUMMI's success was threefold: First, they were given a well-designed product, the Chevy Nova, to build. The Nova would be a version of the Toyota Corolla, a car Toyota was already making and selling in Japan. It would sell for about \$7,500. Second, they understood that the level of technology in a plant was not the primary factor in determining the quality or cost-effectiveness of a product. Rather, the good design and the production system were the key. Third, they realized that car-making was a people business, and high productivity had to do with the way people were organized and managed. NUMMI employed management techniques that were based on the assumption that workers instinctively wanted to do a good job. They played to that instinct, driving out much of the hostility and fear. According to MIT's John Krafchik, since the troublemakers had been weeded out, NUMMI also had the advantage of a well-trained work force, eager for jobs after being laid off for a few years.

Toyota laid the groundwork for success from the outset, by taking time to set up the plant properly, and by not complicating the task of training a new work force with introducing brand-new technology. They selected suppliers and got them involved from the start in effective quality control. They developed the transportation systems to coordinate shipments from Japan and in the United States. They worked out a relationship with the UAW, carefully establishing credibility and trust with the bargaining unit.

At NUMMI, the Toyota Production System was implemented. Its operating philosophy included seven points:

1. *Kaizen*, the never-ending search for perfection (continuous improvement)
2. *Kanban*, the reduction of costs, through its "just-in-time" system

3. Development of full human potential
4. Building mutual trust
5. Developing team performance
6. Treating every employee as manager
7. Providing a stable livelihood for all employees

The striking thing about the seven points was their emphasis on human development. Nowhere did they say, for example, "the search for technological superiority." Within the framework of the seven points it was assumed that technology would or would not be included depending on its ability to enhance quality.

At the core of the human development philosophy were two premises—both new for GM. The first was that the average worker is motivated by the desire to do a job that enhances his sense of self-worth and that earns the respect of other workers. The second premise was that the worker is inspired by an employer who places value in the worker's input. Under the system developed in Japan, every worker is encouraged to think—to use brainpower and find ways of improving products and processes and eliminating waste—and is rewarded for improvements.

At NUMMI, job classifications were reduced to 4 (as opposed to as many as 183 at some GM plants) and employees were placed in teams headed by fellow hourly workers trained in Japan. If a problem was spotted, the team had permission to stop the line. The team leader functioned, not as a boss, but more as a mediator and instructor. Ultimately, each member of the team would be trained to do the leader's job when called upon. More than a year after production was under way, morale was high and quality was high. GM's surveys ranked the Nova regularly among the company's top three models in terms of quality. In fact, only three months after the first cars rolled out of NUMMI, they were already receiving 140 ratings—virtually no defects.

Worker productivity was also rated high. One study suggested that compared with NUMMI, it would probably take about 50 percent more hourly and salaried workers at the old Fremont plant, with comparable technology, to assemble the same car. The same study showed that increased productivity from better management-labor relations reduced the cost of

building each car by about \$750, compared with the traditional GM manufacturing system.

The hiring process itself was different. For one thing, the UAW and NUMMI management collaborated in personnel decisions, selectively hiring from the ranks of workers who had lost their jobs when the Fremont plant closed in 1982. The evaluation process, which included assessment by UAW and NUMMI representatives, took about thirty-five hours for each candidate. By the time a worker was hired, he knew that something special was happening at this plant. The goals of the system were spelled out in great detail, and the importance of *quality* was stressed, above all. This was quite a departure from the lack of orientation that was typical in GM plants at the time.

At NUMMI, there was great attention to detail. Space utilization, for example, was a major concern. The way space was planned reflected concern with thousands of seemingly insignificant details that, when taken together, went far to save money and maintain quality. At the old pre-NUMMI Fremont plant, 12 percent of the factory space was dedicated to reworking defective vehicles, compared with 7 percent at NUMMI, and only 1 percent at NUMMI's Takaoka sister plant in Japan.

Another system that enhanced quality and saved space was the "just-in-time" system, long used in Japanese car companies. Just-in-time means pacing deliveries from suppliers so that plants aren't storing more than a few hours' or a few days' supply of any given part. This system not only saves space and increases plant capacity, it also serves as a quality-control system, since plants don't have huge stocks available to cushion the impact of a high percentage of defects. Even though Fremont had much unutilized floor space, Toyota kept the work space tight to impress upon workers the essence of their philosophy of eliminating waste.

Work standardization amounted to a form of factory-floor choreography. Each team of workers had the responsibility for coordinating body movement, parts location, and time required to do its job. Once the team standardized its work, it found ways to constantly improve upon it. This was called *kaizen*—an ongoing attention to finding low-cost, incremental ways to improve productivity and quality. A second concept, *muda*, was the elimination of waste. At NUMMI, teams were encouraged to docu-

ment any new efficiencies they discovered. Soon workers were using words like *kaizen* and *muda* as commonly as they once might have used "line speedup" and "grievance."

It was not only the hourly workers who were shocked by NUMMI; the salaried staff also found the environment very different. Stephen Bera was one of a group of sixteen GM executives assigned to NUMMI. Bera's area of expertise was production material control. He had been with GM for twenty years, working for Chevrolet, the AC Spark Plug division, and in central staff before joining NUMMI.

Bera thrived at NUMMI. "I went through a personality change out there," he admitted. "I felt better about working for Toyota. No more politics. No more doing things just for my own personal gain. The issue was, 'What's good for the company?' It was one hundred eighty degrees from everything I have ever done in my career." Until NUMMI, Bera said, his motivations were typical: "Do what is good for you as an individual . . . for promotions, for bonuses."

According to Bera, every member of the sixteen-member GM team went through a similar personal transformation. "I think everyone who went out there finally realized that it does no good for one person or one department to succeed. It's either the whole company succeeds or there's no benefit or gain."

This was not the lesson GM necessarily expected to learn from NUMMI. The technological solutions were expected to be more important than changing relationships among people. Bera recalled that he and the other group members were told, "'Go to the joint venture and bring back this *magic* that exists in the Toyota production system.' So everyone went out there with just that in mind: 'Learn what you can. Learn your lessons well. You are going to help us alter the course of the company.'"

"After the first few months, representatives from the corporation came out to California, sat with us, and basically said, 'We're looking to the future from you guys. How does it look to you right now?' Everyone said it was a fascinating experience—one that we had never encountered anywhere else in our careers. We gained so much, we learned our lessons well, and we reported back to the corporation through the personnel group."

Smith and McDonald visited too, eager to get the lowdown

on the NUMMI experience. "We had all the heads of state," Bera said. "They came to talk to us, to tell us, 'Give it your best shot, guys.'"

Of course, the team was already doing that. But the team had other concerns, as well—such as, how was the corporation geared for its repatriation? It was well and good to assign them for two years or so to learn the Japanese methodology, but then what?

"There was no answer to that question for the two years I was out there," Bera said. "No one could answer that question—no one could respond to each and every one of us and tell us just exactly how we were going to go back and give this new-found knowledge and experience to the corporation."

It wasn't that there was *no* corporate plan to incorporate the wisdom of NUMMI into the main body of the company. A liaison office was established near NUMMI whose job it was to make NUMMI techniques available to GM personnel. Videotape presentations and written materials describing NUMMI's production methods were produced, and tours of the NUMMI plant were given. But simply exposing GM personnel to NUMMI was not particularly meaningful from the standpoint of applying the NUMMI systems elsewhere in the company. For one thing, this type of exposure was too superficial to make a difference. More important, the Japanese-style business systems could hardly be implemented by random individuals throughout the company.

A 1986 report by James P. Womack, research associate with the Center for Technology, Policy and Industrial Development at MIT, made the point this way:

"... plant tours, videotapes and manuals seem to be a poor substitute for hands-on experience. Recent interviews by the author with several manufacturing managers in the GM system who have been exposed to NUMMI instructional materials and taken on NUMMI tours reveal a very partial understanding of NUMMI's procedures, and in many cases a defensive conviction that NUMMI techniques can never work in their plant. In the words of one manager, 'All that NUMMI talk is pretty unpopular around here.'"

On the reverse side, the NUMMI employees felt that Toyota was deeply interested in the venture, and they received a lot of attention from top Toyota executives.

"The few times GM executives visited the plant, they did a

'five-minute fly-by,' " said one former NUMMI manager. "So no one believed they were really interested. GM didn't position itself to get maximum benefit from NUMMI because it lacked patience. GM wanted to learn everything in a day—that's a very American character trait. The same wasn't true for Toyota, and they seemed to get much more out of the venture."

In all likelihood, the question of how this learning experience would actually be applied was never considered. For one thing, it was expected that the answer would be technology, rather than people-oriented. It would have been so much easier if Bera and his fellow team members had returned to describe new equipment or production techniques. But, in fact, what the NUMMI team learned concerned a change in management philosophy, and company executives were no doubt reluctant to pursue this direction, for it touched the heart of what was culturally wrong with GM, and this could not be changed by one small experiment in a California town.

It is interesting to reflect about the direction GM was actually taking at the time. Consider the reorganization. It was a radical move, to be sure. But when all was said and done, the reorganization only addressed the issue of *bigness*. It did not shift the paradigm. NUMMI did. It was as though, having reached the conclusion that things didn't work (a giant step, to be sure), the company couldn't quite bring itself to go the next step—except in "experimental" projects like NUMMI or Saturn.

A former NUMMI executive expressed disappointment in how reluctant the company was to deploy the NUMMI team back into the corporation to spread their knowledge. In typical GM fashion, there was no place for them to return to. And so much animosity had developed toward NUMMI that any idea carrying its signature would never have been implemented. In fact, only one member of the team, Bob Hendry, a finance man, would move directly from NUMMI to run a production plant, and it was one of GM's most notorious plants in England. Hendry successfully changed the environment in that plant. Some left the company, others were absorbed into positions that didn't necessarily make use of what they had learned. In any case, the NUMMI "graduates" were merely tokens in the company. Given the system of rotating managers through the plant, at most only fifty managers could be NUMMI-trained by 1992. It appeared that the company wasn't prepared to learn organizational or

people-oriented practices. It expected to learn mechanical or technological lessons.

It wasn't a deliberate lack of commitment. More likely, the implications had never been considered. Roger Smith billed NUMMI as an "educational experience," but his agenda may have been quite different. He may have (1) just wanted to quickly and inexpensively get a small car into production; (2) assumed that the lesson to be learned from the Japanese would be more tangible and less threatening to the General Motors culture; or (3) never really expected to learn from the Japanese as much as he hoped to shake up the company and force his own people to look at the competitive environment.

If Smith did plan to incorporate Japanese wisdom into the GM plants, his choice of method only reveals just how little understanding there was of the underlying reasons the NUMMI method worked so well. It proved how incredibly naïve American management could be when it came to understanding the foreign competition.

TEAM EXPERIMENT: VAN NUYS, CALIFORNIA

As the success of NUMMI filtered back to headquarters, GM synthesized it into its own version of "team concept," which it decided to try in some of its other plants. It did not, as might have been expected, utilize experienced NUMMI-trained executives for this task; its approach was an Americanized version that mimicked the Japanese wisdom only on the most superficial level.

More troubling still was the way GM instituted the plant experiment. Two plants would be pitted against one another—the one with the most efficiency would remain open and the losing plant would be closed. Van Nuys was up against Norwood, a suburban Cincinnati plant that also built Firebirds and Camaros.

From the start, it appeared that Norwood had a great advantage since it was more centrally located. Van Nuys was saddled with an estimated \$400-per-car penalty for getting parts shipped to and cars shipped from the West Coast to markets east of the Mississippi.

But the plant manager, Ernest Schaefer, wasn't going to let

Van Nuys close down without a fight. A skilled negotiator and well-liked manager, Schaefer sat down with the union representatives and convinced them to join the fight. In this way, the GM version of the team concept was introduced to Van Nuys.

The first thing Schaefer did was send his people up to Fremont to observe the way NUMMI operated. Brian Haun remembers being impressed, but surprised to find an absence of high-tech tools. "I was amazed that they basically used 1950s technology, and they did a heck of a job with it. All the press lines were the same as the older types, except, of course, they were new presses. But it wasn't anything fancy."

What Haun saw was a study in plant efficiency. "The workers knew how to change a press line over in two hours. I watched them. I *studied* how they did it. If they wanted to rush, they could even change it in one hour."

The field trip was useful, but hardly sufficient to give the Van Nuys employees any real practical understanding of how to institute Japanese methods and philosophy in their own plant. And observing NUMMI couldn't accomplish the most important thing—to instill mutual trust and respect between workers and management. Van Nuys taught the lesson that you couldn't just shove a new philosophy down people's throats, without making it an integrated part of the corporate understanding. And you couldn't count on union support unless there was a consensus. Although a team of four NUMMI-based managers traveled to Van Nuys frequently during a period of several months to share what they had learned, there was still skepticism in the Van Nuys ranks. "We tried to act as consultants and preach the gospel of NUMMI," one of the four would say later. "But we weren't very good at it. Our approach was hard-sell, and that was the wrong way to convert these guys."

Ray Ruiz, the Van Nuys union chairman was in favor of the team concept. In his view, this was perhaps the only way the plant could survive the threat of plant closings. But there were hot debates in the union about the concept, and the primary issue was one of trust. What were GM's true motivations, union members asked suspiciously. Was this just a roundabout way of extracting concessions from the union?

Schaefer and Ruiz went to work on the workers, organizing mass meetings on each shift, asking for support from the workers on team concept. Union president Peter Beltran, who was

opposed to the plan and called it "the most dangerous scheme ever conjured up by GM to rob workers of their union," pointed out that it wasn't team concept he was opposed to, it was GM's competition, pitting one plant against another. Argued Beltran, if GM expected a commitment from the workers, fine. Give the workers a commitment in return. "A more appropriate exchange for team concept," said Beltran, "would be a GM guarantee that the GM Van Nuys plant would remain open for at least ten years."

Ruiz had support for the plan among many workers, who saw it as a way to get more involved in the production process. Other workers stood behind Beltran. But one union member observed that "the majority are just resigned to it. They say, 'If GM says give us the team concept or we'll close the plant, why are we pretending we have a choice in the matter?'"

When the union members voted, they approved the team concept, but the vote was close and there was still dissent among the ranks. To make matters worse, Ray Ruiz was transferred to Detroit before the plan could get off the ground, and that gave Beltran new impetus because, with Ruiz leaving, a new chairman would have to be elected. Beltran's opposition to the plan hadn't meant as much when Ruiz was the chairman. The union president is relatively removed from the operation of individual plants. But with Ruiz out of the picture, Beltran decided to take a bold stand against the concept. He announced that he would run for chairman himself. He was determined to undermine the team concept at Van Nuys, no matter what it took.

According to Brian Haun, Beltran's opposition was a severe setback, but in many ways it was understandable. "He was an old-school type of guy who had been through some wildcat strikes," Haun explained. "I'm not sure he was completely wrong to say, 'I don't trust those guys'—GM management. They had not given that much reason to be trustworthy. His attitude was kind of that General Motors acts like they want what is good for people, but they don't give a shit and we need to take care of ourselves."

Until Beltran interfered, team concept was on track at Van Nuys, but the workers weren't really committed to the concept; rather, they resented the way they had been bullied into it with the threat that if they didn't go along, the plant would be closed. Beltran won the election.

"So now all the joint union-management committees that had been putting team concept together were dissolved," Haun said.

Part of the problem with implementing team concept at Van Nuys was that it wasn't ever fully understood as being beneficial to the workers or the company. It was merely billed as a way to increase efficiency—a red flag to union workers who perceived higher efficiency as leading naturally to worker layoffs. In fact, this was a legitimate concern. It was hoped that team concept would decrease head count at Van Nuys by as much as 25 percent.

Beltran was determined that this would not happen. The execution of team concept Beltran-style was a twisted version of the original intention that actually decreased efficiency. For example, before team concept, Haun supervised twenty people and had three relief people. When the concept was implemented under Beltran, teams of six were set up, with the plant hiring extra leaders. But the relief people, who were there to cover for absentees, were retained. The result was that Haun now supervised twenty-six people, not twenty-three. Overall in the body shop, at least ten people were added to the payroll.

Under Beltran, efficiency declined on the line, as well. Workers, who now had permission to halt production on the line when quality problems developed, were stopping the line so often that quality was actually decreased—and fewer cars were going out the door.

In plants where the concept was more fully explained, it wasn't perceived as a hoax on the work force. Rather, it was billed as a valuable competitive tool, a cohesive system where workers depend upon one another to be on time and do the job right. When a worker is accountable to his peers, not to a distant, oppressive supervisor, performance improves, because workers don't want to let the rest of the team down by not showing up for work or doing a shoddy job.

In spite of its poor execution of team concept, the Van Nuys plant won the contest against Norwood. In August 1987, the sixty-four-year-old Ohio facility was shut down and four thousand workers were laid off.

How could Van Nuys be the winner when its cost-per-car was \$400 higher than Norwood's? President Jim McDonald would cite a number of reasons, including the age of the Norwood plant. But the real reason might have been that Norwood's

workers had rejected team concept from the start. It was never implemented in the plant.

On April 21, 1988, Peter Beltran would be fired for absenteeism. GM said its investigation showed that Beltran lied in a February 23 letter when he said he missed thirty-eight days of work because of union business. Union vice president Michael Velasquez, who was also fired for signing the letter with Beltran, asserted that Beltran spent those thirty-eight days at union conferences or on paperwork.

It might be suggested that Van Nuys won the battle, but lost the war, since team concept was such a resounding failure. It wasn't all Beltran's fault. Japanese management techniques couldn't simply be overlaid haphazardly on an existing plant. NUMMI's success with team concept started with first selecting people who possessed some leadership qualities and who were open to working in teams. In most General Motors plants, a large percentage of the work force consisted of workers who had been in place for twenty or twenty-five years, and were more resistant to change. Ernie Schaeffer would eventually resolve the heated UAW debate. His successor at Van Nuys, Robert "Bob" Stramy, would bring about further improvements in cost savings and quality. But GM might have avoided many of the problems had it chosen to introduce team concept in a more legitimate way.

In setting up its team concept competition, GM once more fell victim to its habit of oversimplification. If putting workers into teams was the way to succeed, then put the workers into teams. But the team structure was only part of a much larger equation that included product, manufacturing system, plant layout, training, and company philosophy.

GM also ignored one of the most important lessons of all—the need to develop mutual trust between management and workers. Resentment was there from the start; the message that filtered down to the workers was that team concept was a sneaky maneuver to cut back on workers and, ultimately, to close plants. This was no way to establish trust! In pitting plants against one another for survival, General Motors actually may have created the reverse of its intended result—a more militant work force. The savvy union leadership in Van Nuys, for example, openly scorned the idea that in winning the competition they had really won anything at all—except time. What guarantee was there that

the plant would stay open? In fact, it appears that Van Nuys will almost certainly be closed by the end of the decade, as the company deals with long-term issues of declining market share, excess capacity, and slow sales of Camaro and Firebird. Its location, with the added cost of shipping, simply doesn't make economic sense.

One of NUMMI's top executives—a veteran GMer—ranked the three most important lessons learned at NUMMI in this order:

1. Establish good people relationships. This, not advanced technology, was the first priority.
2. Recognize the value of a no-layoff policy. At NUMMI, this policy resulted in worker loyalty and increased productivity. The GM plants lacked such a policy, and workers were suspicious of management's motives.
3. Pay attention to details. It was always on the details that Toyota won. GM paid more attention to the gross investment than it did to the small but all-important details, such as program timing.

Another executive, who was among the first group sent to NUMMI, summarized the lessons learned this way: "You have to create the right environment before you can accomplish anything. GM hasn't always understood the importance of establishing a nonconfrontational relationship with the UAW—it's doing better now, and I think that's because of what we learned at NUMMI. The Japanese system itself, with its constant follow-up on details, and ongoing improvements, helped create an environment where problems could get solved without confrontations. Contrary to the suspicions of many GM people, the Japanese system wasn't mysterious at all. It was simple and basic."

He noted that, in the beginning, the UAW was fearful of NUMMI. "They didn't really understand the concept, and many people thought it sounded like putting the fox in the chicken coop. But now the UAW has come around—they see how beneficial it's been for the workers. The UAW was impressed that, when sales of the Nova were weak, no one got laid off."

In October 1985, the General Motors in-house publication,

GM Today, spotlighted two workers who were rehired at the Fremont plant when NUMMI took over its operation. They were Rick Madrid and Santos Martinez. As team leader, Madrid was sent to Japan for training. "I was part of the media blitz," he said. "Management told me to talk to the press and explain what things were like. They put complete trust in me. Because of that, trust is now mutual."

Martinez echoed the sentiment. "I learned a different meaning for the word respect—one that doesn't include fear," he said. "My responsibility now is to the team, which works together like a family to solve problems and do the job. And no one places blame when something goes wrong."

Madrid and Martinez were sold on the team concept. Their transformation as workers was the direct result of the management practices employed at NUMMI. But on the Fourteenth Floor, far removed from the factories and showrooms, sat long-time GM executives who still could not trust the evidence. And so, the company continued to pay lip service to NUMMI, with plant tours and instructional videos, but it did not move aggressively to package the lessons learned at NUMMI for application in all of its plants.

It seemed that what GM was about was spinning plates in the air. There were several programs going on simultaneously, each with the mission of making the company more competitive, but there was a lack of coordination among the many efforts, and no central system for evaluating them. Was NUMMI the answer? The CPC system? BOC? Spending more on technology? Saturn? In one executive meeting, the question was raised, "Now that twenty-five studies of NUMMI have been completed and a thousand GM people have gone through NUMMI, can we say what the secret of NUMMI is and introduce it to the rest of the company?" The answer was lost in a tremendous debate that led to no conclusion, except that no one on the Executive Committee seemed to have the will to institutionalize change.

To this day, there are those within General Motors who have closed their minds and ears to NUMMI, taking a stance of "I don't want to hear it; it's a Japanese system of thought." Al Warren says this is true, "but we've done a pretty good job evangelizing about NUMMI, and there's a lot more willingness to listen than there used to be."

Resistance has also come from some old-guard managers

and unionists who view the prospect of management-labor cooperation with dread. Some union officials still complain that getting into bed with management will weaken the unions and cost jobs. In their book, *Choosing Sides: Unions and the Team Concept*, former autoworkers Mike Parker and Jane Slaughter railed against the team concept, calling it "management by stress"—a system that was built upon regimentation of workers and job insecurity.

But according to Bruce Lee, director of the United Automobile Workers' western region, that argument, and most other complaints, represent uninformed views. Lee noted in a 1988 *New York Times* article that there is no record of Parker or Slaughter ever stepping foot inside NUMMI, and that their viewpoint is based on fallacy. Wrote Lee, "NUMMI is now at the leading edge of a sea change taking place in American factory management. Decades of managerial arrogance and disregard for the consumer cost American industry hundreds of plants and millions of jobs. Today, at the eleventh hour, progressive companies pushed by progressive unions are finally beginning to put those disastrous mistakes right."

One reason it may be so difficult for some American workers to accept the Japanese labor ideology is that the concept of "continuous improvement" is anathema to their fundamental ideas of what it means to be a productive worker. In the American ideal, the payoff for developing more efficient systems is that the work becomes less demanding. In the Japanese system, the successful solving of one problem simply opens up a new, more difficult, challenge. What some NUMMI workers have termed "management by stress" is really an operating philosophy that never allows workers to stop striving for betterment.

In spite of its struggle to understand and incorporate the NUMMI learning experience, GM has gradually started to come around. Increasingly, the NUMMI-trained managers are being relocated to factories where they can transfer what they've learned. And with the wisdom of hindsight, one former NUMMI manager suggested that it may have been a good thing that the company didn't try to use the NUMMI graduates too forcefully in the beginning.

"We were like religious zealots," he laughed. "We wanted to spread our message to all of GM, but we weren't very sensitive to the concerns of the unconverted. It's no wonder that people

got hostile. NUMMI graduates would go into their plants and act superior to everyone else. In some plants, they'd even form little clubs. We'd try to force the NUMMI concepts down people's throats and tell them how much they had to change. It was ineffective. It's not like that anymore, because more people are knowledgeable about it, and the company is more accepting."

The NUMMI joint venture has a legal cutoff date of 1996, but neither GM nor Toyota has made specific plans for what will happen to the work force. "I have to confess I haven't thought of it," Al Warren hedges. "Time goes by more quickly than you think it does. That's a tremendous work force and they have done a tremendous job. They are now a reservoir of a process that maybe we don't have anymore, and I think we have to look at that."

Now that Toyota has decided to build trucks at NUMMI, the possibility has arisen that Toyota will take over the entire plant when production of the current product, the Prizm, is completed.